

T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after F

This stage has active field content $\text{SM} + S_1 + S_2$. The report combines the renormalisable RGE running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out F . Throughout, $16\pi^2 \mu dX/d\mu = \beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q}Y_u\tilde{H}u + \text{h.c.}$
Y_d	$-\bar{Q}Y_dHd + \text{h.c.}$
Y_e	$-\bar{L}Y_eHe + \text{h.c.}$
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F}M_FF$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 HHS_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$

$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TC L_j)HH$
R_W	$16\pi^2\beta_{C_5}\supset R_W\lambda_5C_{LLS_1S_2}$

Renormalisable EFT1 couplings

$\beta_{g_1}^{(1)}$

Quantity: g_1

$16\pi^2\,\mu\frac{dg_1}{d\mu}=\beta_{g_1}^{(1)}$

Column Representative term structure	
T1	$\frac{43g_1^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{43g_1^3}{6}$

$\beta_{g_2}^{(1)}$
Quantity: g_2

$$16\pi^2\,\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{17g_2^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{17g_2^3}{6}$

$\beta_{g_3}^{(1)}$
Quantity: g_3

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7g_3^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-7g_3^3$

$\beta_{Y_u}^{(1)}$

Quantity: Y_u

$$16\pi^2\,\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$
T2	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$
T3	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$
T3	$-8g_3^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

Quantity: Y_d

$$16\pi^2\,\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

Quantity: Y_e

$$16\pi^2\,\mu\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$
T3	$+\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$
T2	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$
T3	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$

$\beta_{m_1^2}^{(1)}$

Quantity: m_1^2

$16\pi^2\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$-\frac{9g_2^2m_1^2}{2}$
T2	$-\frac{3g_1^2m_1^2}{2}$
T3	$+4\lambda_{12}^{(1)}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9g_2^2m_1^2}{2}$	$-\frac{3g_1^2m_1^2}{2}$	$+4\lambda_{12}^{(1)}m_2^2$

Quantity: m_1^2

Terms 2 of 2

Column	Representative term structure
T1	$+6\lambda_{S_1}^{(1)}m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{S_1}^{(1)}m_1^2$

$\beta_{m_2^2}^{(1)}$

Quantity: m_2^2

$16\pi^2\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$-\frac{9g_2^2m_2^2}{2}$
T2	$-\frac{3g_1^2m_2^2}{2}$
T3	$+4\lambda_{12}^{(1)}m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$	$+4\lambda_{12}^{(1)}m_1^2$

Quantity: m_2^2

Terms 2 of 2

Column	Representative term structure
T1	$+6\lambda_{S_2}^{(1)}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{S_2}^{(1)}m_2^2$

$\beta_{\lambda_1}^{(1)}$

Quantity: λ_1

$$16\pi^2\,\mu\,\frac{d\lambda_1}{d\mu}=\beta_{\lambda_1}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$
T3	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$

Quantity: λ_1

Terms 2 of 6

Column	Representative term structure
T1	$-12\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\,Y_u\,(Y_u)^\dagger\Big)$
T2	$+8\lambda_7\,(\lambda_7)^*$
T3	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-12\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\,Y_u\,(Y_u)^\dagger\Big)$	$+8\lambda_7\,(\lambda_7)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\right)^*$

Quantity: λ_1

Terms 3 of 6

Column	Representative term structure
T1	$+8\lambda_8\left(\lambda_8\right)^*$
T2	$+4\lambda_5\left(\lambda_5\right)^*$
T3	$+ \frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_8\left(\lambda_8\right)^*$	$+4\lambda_5\left(\lambda_5\right)^*$	$+ \frac{9g_2^4}{4}$

Quantity: λ_1

Terms 4 of 6

Column	Representative term structure
T1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$
T2	$-9g_2^2\lambda_1$
T3	$+\frac{3g_1^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$

Quantity: λ_1

Terms 5 of 6

Column	Representative term structure
T1	$+12\lambda_1^2$
T2	$+4\lambda_{HS_1}^{(1)2}$
T3	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$

Quantity: λ_1

Terms 6 of 6

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)^2}$
T2	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{HS_2}^{(1)^2}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$

$$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$$

Quantity: $\lambda_{S_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$8\lambda_7\left(\lambda_7\right)^*$
T2	$+8\lambda_9\left(\lambda_9\right)^*$
T3	$+24\lambda_{11}\left(\lambda_{11}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$8\lambda_7\left(\lambda_7\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$	$+24\lambda_{11}\left(\lambda_{11}\right)^*$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+ \frac{9 g_2^4}{4}$
T2	$+ g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_{S_1}^{(1)} \right)$
T3	$- 9 g_2^2 \lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{9 g_2^4}{4}$	$+ g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_{S_1}^{(1)} \right)$	$- 9 g_2^2 \lambda_{S_1}^{(1)}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+ \frac{3g_1^4}{4}$
T2	$+4\lambda_{12}^{(1)2}$
T3	$+ \frac{\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{3g_1^4}{4}$	$+4\lambda_{12}^{(1)2}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1) \ 2}$
T2	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$
T3	$+12\lambda_{S_1}^{(1) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{HS_1}^{(1) \ 2}$	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	$+12\lambda_{S_1}^{(1) \ 2}$

$\beta_{\lambda_{S_2}}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{(1)}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$8\lambda_8\left(\lambda_8\right)^*$
T2	$+8\lambda_9\left(\lambda_9\right)^*$
T3	$+24\lambda_{10}\left(\lambda_{10}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$8\lambda_8\left(\lambda_8\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$	$+24\lambda_{10}\left(\lambda_{10}\right)^*$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+ \frac{9 g_2^4}{4}$
T2	$+ g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_{S_2}^{(1)} \right)$
T3	$- 9 g_2^2 \lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{9 g_2^4}{4}$	$+ g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_{S_2}^{(1)} \right)$	$- 9 g_2^2 \lambda_{S_2}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+ \frac{3g_1^4}{4}$
T2	$+4\lambda_{12}^{(1)2}$
T3	$+ \frac{\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{3g_1^4}{4}$	$+4\lambda_{12}^{(1)2}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1) \ 2}$
T2	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$
T3	$+12\lambda_{S_2}^{(1) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	$+12\lambda_{S_2}^{(1) \ 2}$

$$\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Quantity: $\lambda_{HS_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(1)}}{d\mu}=\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{HS_1}^{(1)}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$+24\lambda_7\,(\lambda_7)^*$
T3	$+3\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{HS_1}^{(1)}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$+24\lambda_7\,(\lambda_7)^*$	$+3\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 2 of 5

Column	Representative term structure
T1	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 (\lambda_{11})^*)$
T2	$+ 6 \lambda_5 (\lambda_5)^*$
T3	$+ \frac{9 g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 (\lambda_{11})^*)$	$+ 6 \lambda_5 (\lambda_5)^*$	$+ \frac{9 g_2^4}{4}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 5

Column	Representative term structure
T1	$-9g_2^2\lambda_{HS_1}^{(1)}$
T2	$+\frac{3g_1^4}{4}$
T3	$-3g_1^2\lambda_{HS_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{HS_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T2	$+6\lambda_1\lambda_{HS_1}^{(1)}$
T3	$+4\lambda_{HS_1}^{(1)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 5 of 5

Column	Representative term structure
T1	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T2	$+\frac{3\lambda_{HS_1}^{(A) \ 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3\lambda_{HS_1}^{(A) \ 2}}{4}$

$\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$

Quantity: $\lambda_{HS_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(1)}}{d\mu}=\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{HS_2}^{(1)}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+3\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{10}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{HS_2}^{(1)}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+3\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{10}\right)^*)$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 2 of 5

Column	Representative term structure
T1	$+24\lambda_8\left(\lambda_8\right)^*$
T2	$+6\lambda_5\left(\lambda_5\right)^*$
T3	$+ \frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+24\lambda_8\left(\lambda_8\right)^*$	$+6\lambda_5\left(\lambda_5\right)^*$	$+ \frac{9g_2^4}{4}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 3 of 5

Column	Representative term structure
T1	$-9g_2^{-2}\lambda_{HS_2}^{(1)}$
T2	$+\frac{3g_1^4}{4}$
T3	$-3g_1^{-2}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^{-2}\lambda_{HS_2}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^{-2}\lambda_{HS_2}^{(1)}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$
T2	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3	$+4\lambda_{HS_2}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 5 of 5

Column	Representative term structure
T1	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T2	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{4}$

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}^{(1)}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$+24\lambda_9\left(\lambda_9\right)^*$
T3	$+9\lambda_{10}\left((\lambda_{10})^*+(\lambda_{11})^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+24\lambda_9\left(\lambda_9\right)^*$	$+9\lambda_{10}\left((\lambda_{10})^*+(\lambda_{11})^*\right)$ $+9\lambda_{11}\left((\lambda_{10})^*+(\lambda_{11})^*\right)$

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 5

Column	Representative term structure
T1	$+3\lambda_5\left(\lambda_5\right)^*$
T2	$+\frac{9g_2^4}{4}$
T3	$-9g_2^2\lambda_{12}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+3\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{12}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 5

Column	Representative term structure
T1	$+ \frac{3g_1^4}{4}$
T2	$-3g_1^2\lambda_{12}^{(1)}$
T3	$+4\lambda_{12}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 5

Column	Representative term structure
T1	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T2	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3	$+\frac{3\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$

Quantity: $\lambda_{12}^{(1)}$

Terms 5 of 5

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$\beta_{\lambda_5}^{(1)}$
Quantity: λ_5

$$16\pi^2\,\mu\,\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_5\left(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$
T2	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_7\right)^*$
T3	$+4\lambda_{11}\left(\lambda_7\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_5\left(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_7\right)^*$	$+4\lambda_{11}\left(\lambda_7\right)^*$

Quantity: λ_5

Terms 2 of 4

Column	Representative term structure
T1	$+4\left(\lambda_8\right)^*\left(\left(\lambda_{10}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$
T2	$-9g_2^2\lambda_5$
T3	$-3g_1^2\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\left(\lambda_8\right)^*\left(\left(\lambda_{10}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	$-9g_2^2\lambda_5$	$-3g_1^2\lambda_5$

Quantity: λ_5

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_5$
T2	$+2\lambda_1\lambda_5$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Quantity: λ_5

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{HS_2}^{(1)}\lambda_5$

$\beta_{\lambda_{12}^{(A)}}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(A)}}{d\mu}=\beta_{\lambda_{12}^{(A)}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$
T2	$+\frac{4}{3}\lambda_{10}\left(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*\right)$
T3	$+\frac{4}{3}\lambda_{11}\left(\left(\lambda_{10}\right)^*+5\left(\lambda_{11}\right)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$	$+\frac{4}{3}\lambda_{10}\left(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*\right)$	$+\frac{4}{3}\lambda_{11}\left(\left(\lambda_{10}\right)^*+5\left(\lambda_{11}\right)^*\right)$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$
T2	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$
T3	$-9g_2^2\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T2	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$
T3	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

$$\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Quantity: $\lambda_{HS_1}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(A)}}{d\mu}=\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{HS_1}^{(A)}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\right)$
T2	$+\frac{64}{3}\lambda_7\left(\lambda_7\right)^*$
T3	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{HS_1}^{(A)}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\right)$	$+\frac{64}{3}\lambda_7\left(\lambda_7\right)^*$	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*)$
T2	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$
T3	$+g_1^2\left(-6g_2^2-3\lambda_{HS_1}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*)$	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{HS_1}^{(A)}\right)$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_{HS_1}^{(A)}$
T2	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3	$+2\lambda_1\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 4 of 4

Column	Representative term structure
T1	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T2	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

$\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$

Quantity: $\lambda_{HS_2}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(A)}}{d\mu}=\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{HS_2}^{(A)}(6\text{Tr}\big(Y_d\;(Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\;(Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\;(Y_u)^\dagger\big))$
T2	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{10})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{HS_2}^{(A)}(6\text{Tr}\big(Y_d\;(Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\;(Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\;(Y_u)^\dagger\big))$	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{10})^*)$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 4

Column	Representative term structure
T1	$+64\lambda_8\left(\lambda_8\right)^*$
T2	$+16\lambda_5\left(\lambda_5\right)^*$
T3	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+64\lambda_8\left(\lambda_8\right)^*$	$+16\lambda_5\left(\lambda_5\right)^*$	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_{HS_2}^{(A)}$
T2	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 4 of 4

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T2	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$

$\beta_{\lambda_7}^{(1)}$
Quantity: λ_7

$$16\pi^2\,\mu\,\frac{d\lambda_7}{d\mu}=\beta_{\lambda_7}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_7(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$
T3	$+2\lambda_{11}(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_7(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$	$+2\lambda_{11}(\lambda_5)^*$

Quantity: λ_7

Terms 2 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_7$
T2	$-3g_1^2\lambda_7$
T3	$+2\lambda_1\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_7$	$-3g_1^2\lambda_7$	$+2\lambda_1\lambda_7$

Quantity: λ_7

Terms 3 of 4

Column	Representative term structure
T1	$+8\lambda_{HS_1}^{(1)}\lambda_7$
T2	$+\frac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$
T3	$+2\lambda_7\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{HS_1}^{(1)}\lambda_7$	$+\frac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$	$+2\lambda_7\lambda_{S_1}^{(1)}$

Quantity: λ_7

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_8\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_8\lambda_9$

$\beta_{\lambda_{12}}^{(1)}$

Quantity: λ_{12}

$$16\pi^2\,\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+6\lambda_9\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+6\lambda_9\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$

Quantity: λ_{12}

Terms 2 of 4

Column	Representative term structure	
T1	$-9g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	
T2	$-3g_1^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	
T3	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$-3g_1^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$

Terms 3 of 4

Column	Representative term structure
T1	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$+\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11})$
T3	$+\lambda_{HS_1}^{(A)}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11})$	$+\lambda_{HS_1}^{(A)}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10})$
T2	$+\lambda_{HS_2}^{(A)}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$
T3	$-6\lambda_7\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10})$	$+\lambda_{HS_2}^{(A)}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$	$-6\lambda_7\lambda_5$

$\beta_{\lambda_8}^{(1)}$
Quantity: λ_8

$$16\pi^2\,\mu\,\frac{d\lambda_8}{d\mu}=\beta_{\lambda_8}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_8(6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$
T2	$+4\lambda_7\,(\lambda_9)^*$
T3	$+2\,(\lambda_5)^*\,((\lambda_{10})^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_8(6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$	$+4\lambda_7\,(\lambda_9)^*$	$+2\,(\lambda_5)^*\,((\lambda_{10})^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$

Quantity: λ_8

Terms 2 of 3

Column	Representative term structure
T1	$-9g_2^2\lambda_8$
T2	$-3g_1^2\lambda_8$
T3	$+2\lambda_1\lambda_8$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_8$	$-3g_1^2\lambda_8$	$+2\lambda_1\lambda_8$

Quantity: λ_8

Terms 3 of 3

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_8$
T2	$+4\lambda_{HS_2}^{(A)}\lambda_8$
T3	$+2\lambda_8\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$	$+2\lambda_8\lambda_{S_2}^{(1)}$

$\beta_{\lambda_{11}}^{(1)}$

Quantity: λ_{11}

$16\pi^2\mu\frac{d\lambda_{11}}{d\mu}=\beta_{\lambda_{11}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$4\lambda_9((\lambda_{10})^*+5\,(\lambda_{11})^*)$
T2	$-9g_2^2\lambda_{11}$
T3	$-3g_1^2\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$4\lambda_9((\lambda_{10})^*+5\,(\lambda_{11})^*)$	$-9g_2^2\lambda_{11}$	$-3g_1^2\lambda_{11}$

Quantity: λ_{11}

Terms 2 of 3

Column	Representative term structure
T1	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T2	$+\lambda_{12}^{(A)}\left(\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$
T3	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Quantity: λ_{11}

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$
T2	$+4\lambda_7\lambda_5$
T3	$+12\lambda_{S_1}^{(1)}\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_7\lambda_5$	$+12\lambda_{S_1}^{(1)}\lambda_{11}$

$\beta_{\lambda_9}^{(1)}$
Quantity: λ_9

$16\pi^2\,\mu\frac{d\lambda_9}{d\mu}=\beta_{\lambda_9}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$4\lambda_7\left(\lambda_8\right)^*$
T2	$-9g_2^2\lambda_9$
T3	$-3g_1^2\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$4\lambda_7\left(\lambda_8\right)^*$	$-9g_2^2\lambda_9$	$-3g_1^2\lambda_9$

Quantity: λ_9

Terms 2 of 4

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_9$
T2	$+\frac{4\lambda_{12}^{(A)}}{3}\lambda_9$
T3	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{12}^{(1)}\lambda_9$	$+\frac{4\lambda_{12}^{(A)}}{3}\lambda_9$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$

Quantity: λ_9

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_{S_1}^{(1)}\lambda_9$
T2	$+2\lambda_9\lambda_{S_2}^{(1)}$
T3	$+5\lambda_{10}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$	$+5\lambda_{10}^2$ $+5\lambda_{11}^2$

Quantity: λ_9

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_{10}\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{10}\lambda_{11}$

$\beta_{\lambda_{10}}^{(1)}$

Quantity: λ_{10}

$16\pi^2\mu\frac{d\lambda_{10}}{d\mu}=\beta_{\lambda_{10}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$
T2	$+4\lambda_9\left(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*\right)$
T3	$-9g_2^2\lambda_{10}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$+4\lambda_9\left(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*\right)$	$-9g_2^2\lambda_{10}$

Quantity: λ_{10}

Terms 2 of 3

Column	Representative term structure
T1	$-3g_1^2\lambda_{10}$
T2	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T3	$+\lambda_{12}^{(A)}\left(\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_1^2\lambda_{10}$	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$

Quantity: λ_{10}

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$
T3	$+12\lambda_{10}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$+12\lambda_{10}\lambda_{S_2}^{(1)}$

Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis C_{ijab} . Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$$\beta_{C_{1133}}^{(1)}$$

Quantity: 1, 1, 3, 3

$$16\pi^2\mu\frac{dC_{1133}}{d\mu}=\beta_{C_{1133}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1134}}^{(1)}$

Quantity: 1, 1, 3, 4

$$16\pi^2\,_{\mu}\frac{dC_{1134}}{d\mu}=\beta_{C_{1134}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1144}}^{(1)}$

Quantity: 1, 1, 4, 4

$16\pi^2\,_{\mu}\frac{dC_{1144}}{d\mu}=\beta_{C_{1144}}^{(1)}$

Column	Representative term structure
T1	$\frac{lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{2lambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta_{C_{1155}}^{(1)}$

Quantity: 1, 1, 5, 5

$16\pi^2\,\mu\frac{dC_{1155}}{d\mu}=\beta_{C_{1155}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$-\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{12MF}$

$\beta_{C_{1156}}^{(1)}$

Quantity: 1, 1, 5, 6

$$16\pi^2\,\mu\,\frac{dC_{1156}}{d\mu}=\beta_{C_{1156}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{y\overline{e}}}{12MF}$

$\beta_{C_{11511}}^{(1)}$
Quantity: 1, 1, 5, 11

$$16\pi^2\,\mu\frac{dC_{11511}}{d\mu}=\beta_{C_{11511}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambda12Inv2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{lambda12Inv2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{4MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{11512}}^{(1)}$
Quantity: 1, 1, 5, 12

$$16\pi^2\,\mu\frac{dC_{11512}}{d\mu}=\beta_{C_{11512}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{1166}}^{(1)}$

Quantity: 1, 1, 6, 6

$16\pi^2\,\mu\frac{dC_{1166}}{d\mu}=\beta_{C_{1166}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{12MF}$

$\beta_{C_{11611}}^{(1)}$
Quantity: 1, 1, 6, 11

$$16\pi^2\,\mu\frac{dC_{11611}}{d\mu}=\beta_{C_{11611}}^{(1)}$$

Column	Representative term structure
T1	$\frac{i\lambda\,Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ie\overline{y_1y_2}y\overline{e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{i\lambda\,Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ie\overline{y_1y_2}y\overline{e}}{12MF}$

$\beta_{C_{11612}}^{(1)}$
Quantity: 1, 1, 6, 12

$$16\pi^2\,\mu\frac{dC_{11612}}{d\mu}=\beta_{C_{11612}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambda12Inv2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{lambda12Inv2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{4MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$\beta_{C_{111111}}^{(1)}$
Quantity: 1, 1, 11, 11

$$16\pi^2\mu\frac{dC_{111111}}{d\mu}=\beta_{C_{111111}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}lambdaS2Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{12MF}$

$\beta_{C_{1111112}}^{(1)}$

Quantity: 1, 1, 11, 12

$$16\pi^2\,_{\mu}\frac{dC_{1111112}}{d\mu}=\beta_{C_{1111112}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y\overline{e}}}{12MF}$

$\beta_{C_{111212}}^{(1)}$

Quantity: 1, 1, 12, 12

$$16\pi^2{}_{\mu}\frac{dC_{111212}}{d\mu}=\beta_{C_{111212}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda daS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$	$\frac{\sqrt{3}\lambda daS2Inv1\overline{y_2}^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{12MF}$

$$\beta_{C_{1213}}^{(1)}$$

Quantity: 1, 2, 1, 3

$$16\pi^2\,_{\mu}\frac{dC_{1213}}{d\mu}=\beta_{C_{1213}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{lambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta_{C_{1214}}^{(1)}$

Quantity: 1, 2, 1, 4

$$16\pi^2\,_{\mu}\frac{dC_{1214}}{d\mu}=\beta_{C_{1214}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1223}}^{(1)}$

Quantity: 1, 2, 2, 3

$$16\pi^2\,_{\mu}\frac{dC_{1223}}{d\mu}=\beta_{C_{1223}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1224}}^{(1)}$

Quantity: 1, 2, 2, 4

$$16\pi^2\,_{\mu}\frac{dC_{1224}}{d\mu}=\beta_{C_{1224}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\lambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta_{C_{1257}}^{(1)}$

Quantity: 1, 2, 5, 7

$16\pi^2\,\mu\frac{dC_{1257}}{d\mu}=\beta_{C_{1257}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{8MF}$	$-\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{12MF}$	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{24MF}$

$\beta_{C_{1258}}^{(1)}$
Quantity: 1, 2, 5, 8

$$16\pi^2\,\mu\,\frac{dC_{1258}}{d\mu}=\beta_{C_{1258}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{y\overline{e}}}{24MF}$

$\beta_{C_{1259}}^{(1)}$
Quantity: 1, 2, 5, 9

$$16\pi^2\,_{\mu}\frac{dC_{1259}}{d\mu}=\beta_{C_{1259}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{lambda12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$

$\beta_{C_{12510}}^{(1)}$
Quantity: 1, 2, 5, 10

$$16\pi^2\,\mu\frac{dC_{12510}}{d\mu}=\beta_{C_{12510}}^{(1)}$$

Column	Representative term structure
T1	$\frac{i\lambda da12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{i\lambda da12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{24MF}$

$\beta_{C_{1267}}^{(1)}$

Quantity: 1, 2, 6, 7

$$16\pi^2\,\mu\,\frac{dC_{1267}}{d\mu}=\beta_{C_{1267}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{y\overline{e}}}{24MF}$

$\beta_{C_{1268}}^{(1)}$

Quantity: 1, 2, 6, 8

$$16\pi^2\,\mu\,\frac{dC_{1268}}{d\mu}=\beta_{C_{1268}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{8MF}$	$\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{12MF}$	$-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{24MF}$

$\beta_{C_{1269}}^{(1)}$
Quantity: 1, 2, 6, 9

$$16\pi^2\,_{\mu}\frac{dC_{1269}}{d\mu}=\beta_{C_{1269}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$

$\beta_{C_{12610}}^{(1)}$
Quantity: 1, 2, 6, 10

$$16\pi^2\,\mu\frac{dC_{12610}}{d\mu}=\beta_{C_{12610}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{lambda12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$

$\beta_{C_{12711}}^{(1)}$
Quantity: 1, 2, 7, 11

$$16\pi^2\,\mu\frac{dC_{12711}}{d\mu}=\beta_{C_{12711}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$

$\beta_{C_{12712}}^{(1)}$
Quantity: 1, 2, 7, 12

$$16\pi^2\,\mu\frac{dC_{12712}}{d\mu}=\beta_{C_{12712}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{24MF}$

$\beta_{C_{12811}}^{(1)}$
Quantity: 1, 2, 8, 11

$$16\pi^2\,\mu\frac{dC_{12811}}{d\mu}=\beta_{C_{12811}}^{(1)}$$

Column	Representative term structure
T1	$\frac{i\lambda da12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{i\lambda da12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{24MF}$

$\beta_{C_{12812}}^{(1)}$
Quantity: 1, 2, 8, 12

$$16\pi^2\,\mu\frac{dC_{12812}}{d\mu}=\beta_{C_{12812}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{lambda12Inv2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$

$\beta_{C_{12911}}^{(1)}$
Quantity: 1, 2, 9, 11

$$16\pi^2\,\mu\frac{dC_{12911}}{d\mu}=\beta_{C_{12911}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{8MF}$	$\frac{\sqrt{3}lambdaS2Inv1\overline{y_2}^2}{12MF}$	$-\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{24MF}$

$\beta_{C_{12912}}^{(1)}$
Quantity: 1, 2, 9, 12

$$16\pi^2\,\mu\frac{dC_{12912}}{d\mu}=\beta_{C_{12912}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{24MF}$

$\beta_{C_{121011}}^{(1)}$
Quantity: 1, 2, 10, 11

$$16\pi^2\mu\frac{dC_{121011}}{d\mu}=\beta_{C_{121011}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{24MF}$

$\beta_{C_{121012}}^{(1)}$

Quantity: 1, 2, 10, 12

$16\pi^2\mu\frac{dC_{121012}}{d\mu}=\beta_{C_{121012}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_2}^2}{8MF}$	$-\frac{\sqrt{3}lambdaS2Inv1\overline{y_2}^2}{12MF}$	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{24MF}$

$$\beta_{C_{2211}}^{(1)}$$

Quantity: 2, 2, 1, 1

$$16\pi^2\,_{\mu}\frac{dC_{2211}}{d\mu}=\beta_{C_{2211}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2\lambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta_{C_{2212}}^{(1)}$

Quantity: 2, 2, 1, 2

$$16\pi^2\,_{\mu}\frac{dC_{2212}}{d\mu}=\beta_{C_{2212}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$

$$\beta_{C_{2222}}^{(1)}$$

Quantity: 2, 2, 2, 2

$$16\pi^2\,_{\mu}\frac{dC_{2222}}{d\mu}=\beta_{C_{2222}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{2\lambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta_{C_{2277}}^{(1)}$

Quantity: 2, 2, 7, 7

$16\pi^2\,_{\mu}\frac{dC_{2277}}{d\mu}=\beta_{C_{2277}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$-\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{12MF}$

$\beta_{C_{2278}}^{(1)}$

Quantity: 2, 2, 7, 8

$$16\pi^2\,\mu\,\frac{dC_{2278}}{d\mu}=\beta_{C_{2278}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{y\overline{e}}}{12MF}$

$\beta_{C_{2279}}^{(1)}$

Quantity: 2, 2, 7, 9

$16\pi^2\mu\frac{dC_{2279}}{d\mu}=\beta_{C_{2279}}^{(1)}$

Column	Representative term structure
T1	$\frac{lambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{lambda12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{22710}}^{(1)}$
Quantity: 2, 2, 7, 10

$$16\pi^2\,\mu\frac{dC_{22710}}{d\mu}=\beta_{C_{22710}}^{(1)}$$

Column	Representative term structure
T1	$\frac{i\lambda da12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{i\lambda da12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{12MF}$

$\beta_{C_{2288}}^{(1)}$

Quantity: 2, 2, 8, 8

$$16\pi^2\,\mu\,\frac{dC_{2288}}{d\mu}=\beta_{C_{2288}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}lambdaS1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\overline{e}}}{12MF}$

$\beta_{C_{2289}}^{(1)}$

Quantity: 2, 2, 8, 9

$16\pi^2\,_{\mu}\frac{dC_{2289}}{d\mu}=\beta_{C_{2289}}^{(1)}$

Column	Representative term structure
T1	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{22810}}^{(1)}$
Quantity: 2, 2, 8, 10

$$16\pi^2\,\mu\frac{dC_{22810}}{d\mu}=\beta_{C_{22810}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{lambda12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{2299}}^{(1)}$
Quantity: 2, 2, 9, 9

$$16\pi^2\,\mu\frac{dC_{2299}}{d\mu}=\beta_{C_{2299}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}lambdaS2Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{12MF}$

$\beta_{C_{22910}}^{(1)}$
Quantity: 2, 2, 9, 10

$$16\pi^2\,\mu\frac{dC_{22910}}{d\mu}=\beta_{C_{22910}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y\overline{e}}}{12MF}$

$\beta_{C_{221010}}^{(1)}$

Quantity: 2, 2, 10, 10

$$16\pi^2{}_{\mu}\frac{dC_{221010}}{d\mu}=\beta_{C_{221010}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$	$\frac{\sqrt{3}lambdaS2Inv1\overline{y_2}^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{12MF}$